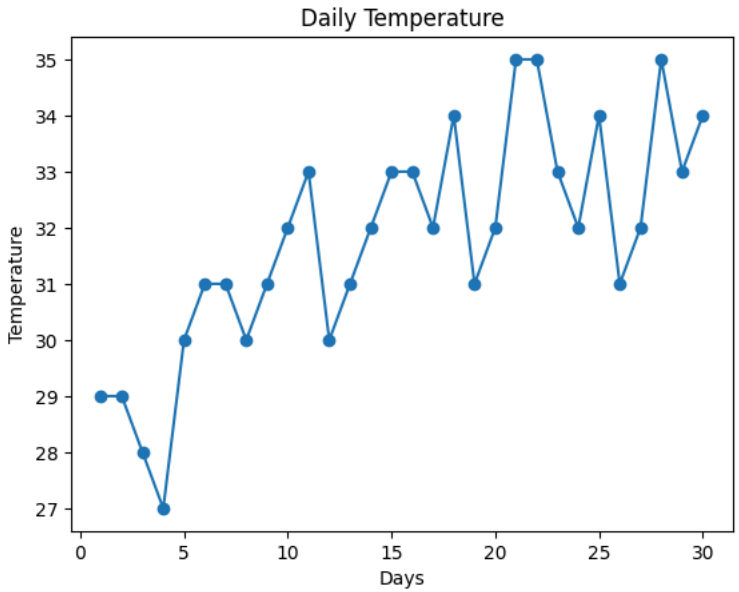
A. Concept Selection Questions (Think before plotting)

1. Given daily temperature data for 30 days, which chart will you use and why?

```
import matplotlib.pyplot as plt
temperature = [29,29,28,27,30,31,31,30,31,32,33,30,31,32,33,33,32,34,31,32,35,
days = range(1,31)
plt.plot(days,temperature, marker='o')
plt.xlabel('Days')
plt.ylabel('Temperature')
plt.title('Daily Temperature')
plt.show()
```

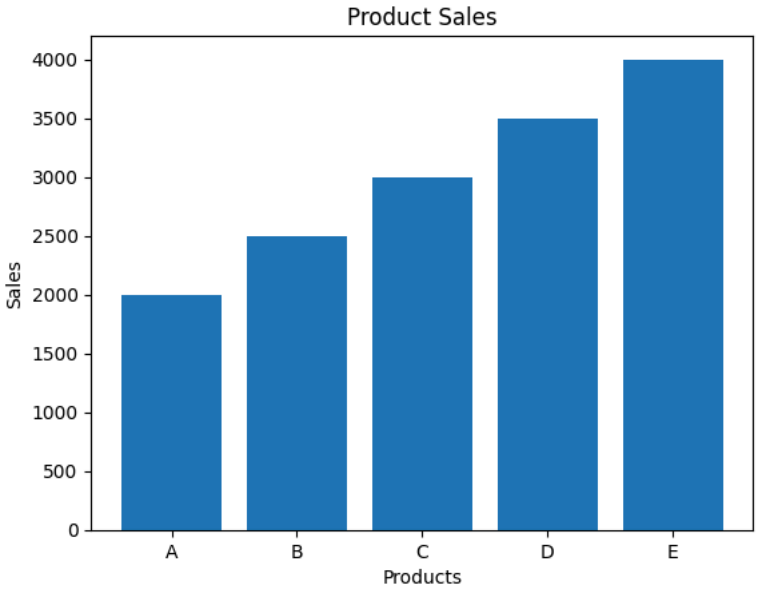


2.

You are comparing sales of 5 different products in a month. Which chart is most suitable?

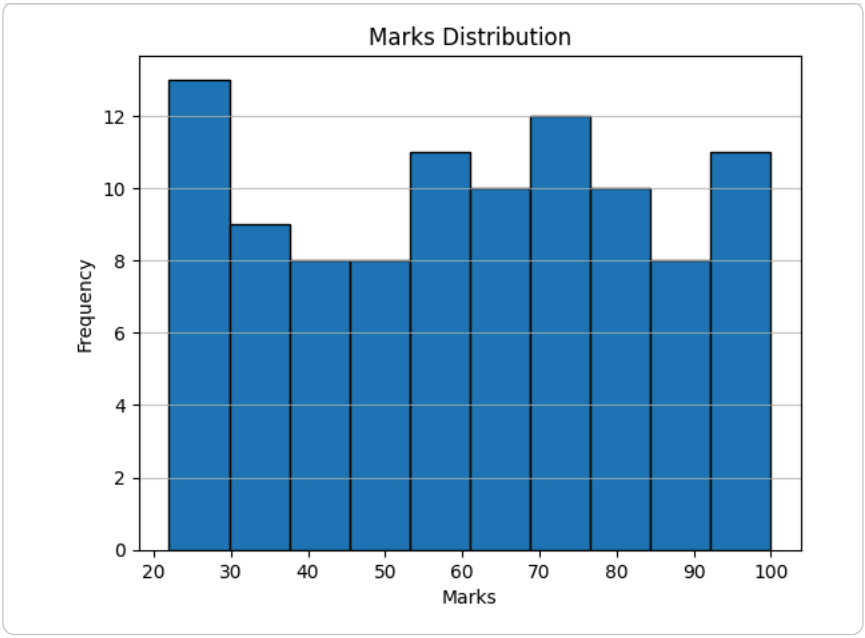
```
# BAR CHART
import matplotlib.pyplot as plt
product = ["A","B", "C", "D", "E"]
sales = [2000,2500,3000,3500,4000]
```

```
plt.bar(product,sales)
plt.title("Product Sales")
plt.xlabel("Products")
plt.ylabel("Sales")
plt.show()
```



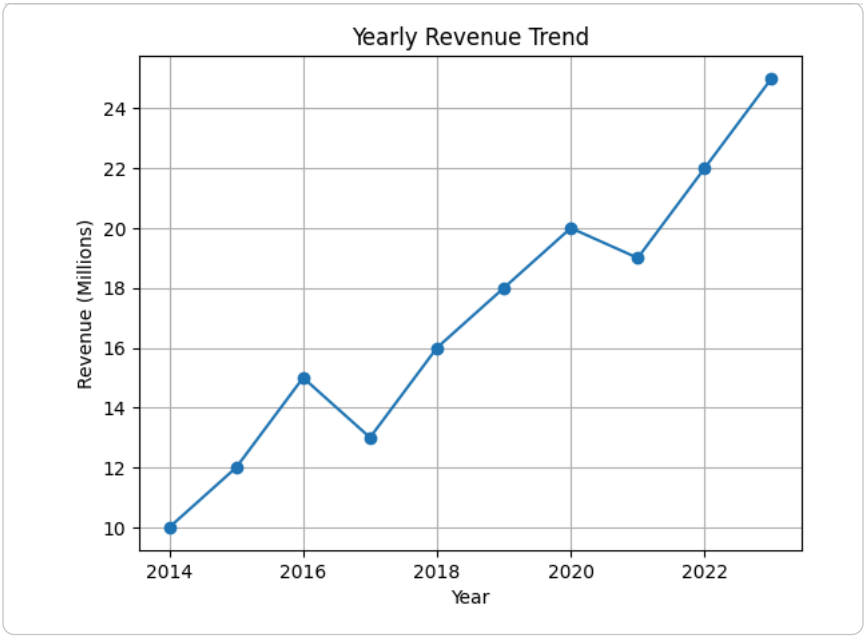
3. Marks of 100 students are given. Which visualization will best show how marks are distributed?

```
import matplotlib.pyplot as plt
import random
marks = [random.randint(20, 100) for _ in range(100)]
plt.hist(marks, bins=10, edgecolor='black')
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.grid(axis='y', alpha=0.75)
plt.show()
```



4. You have yearly revenue data for the last 10 years. Which chart will clearly show the trend?

```
import matplotlib.pyplot as plt
years = range(2014, 2024)
revenue = [10, 12, 15, 13, 16, 18, 20, 19, 22, 25]
plt.plot(years, revenue, marker='o')
plt.title('Yearly Revenue Trend')
plt.xlabel('Year')
plt.ylabel('Revenue (Millions)')
plt.grid(True)
plt.show()
```



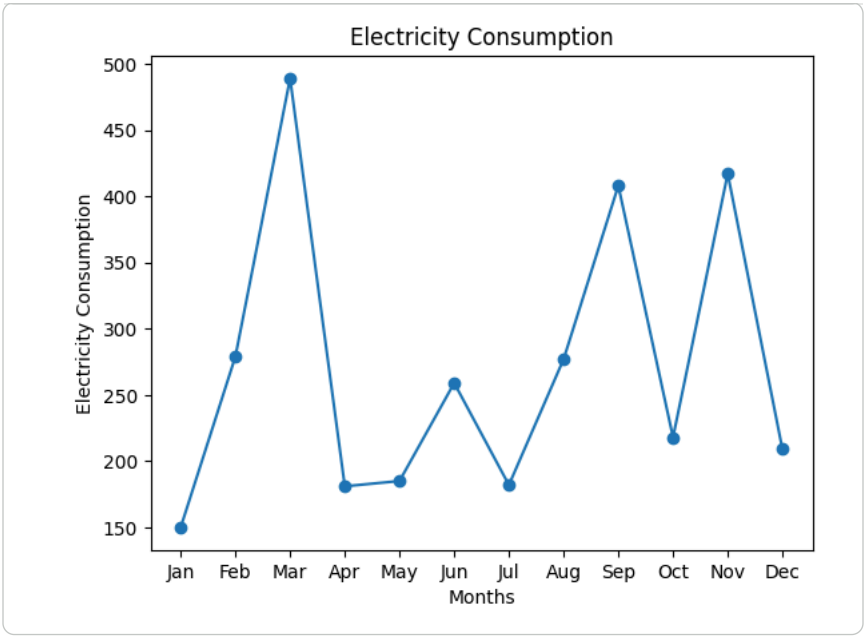
5. Website response times are stored in a list. Which chart will help analyze system performance?

```
#Line chart
```

B. Line Chart Practice Questions

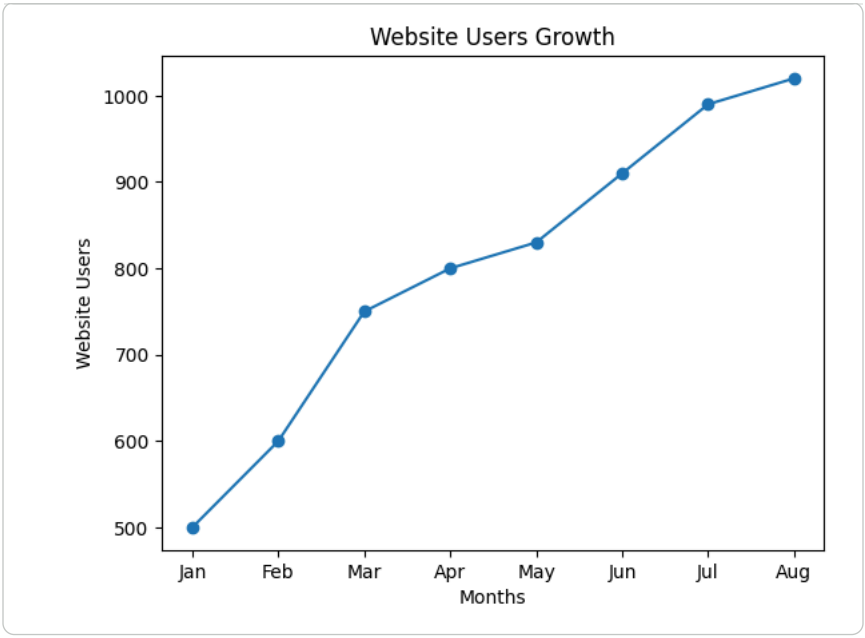
1. Plot a line chart showing monthly electricity consumption for 12 months.

```
import matplotlib.pyplot as plt
import random
electricity = random.sample(range(100,500),12)
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct
plt.plot(months,electricity, marker='o')
plt.title("Electricity Consumption")
plt.xlabel("Months")
plt.ylabel("Electricity Consumption")
plt.show()
```



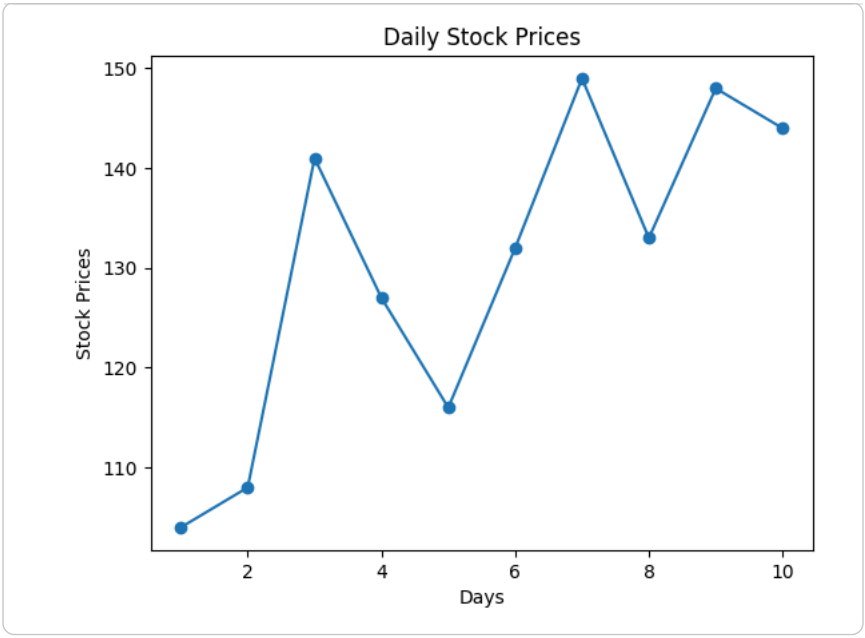
2. Create a line chart to show the growth of website users over 8 months.

```
import matplotlib.pyplot as plt
users = [500,600,750,800,830,910,990,1020]
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug"]
plt.plot(months,users, marker='o')
plt.title("Website Users Growth")
plt.xlabel("Months")
plt.ylabel("Website Users")
plt.show()
```



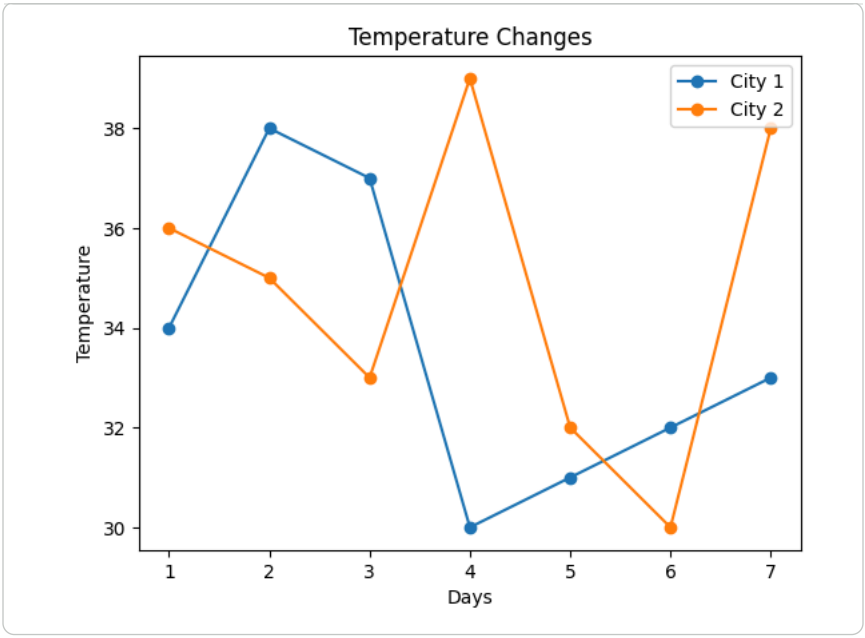
3. Draw a line chart for daily stock prices and identify whether the trend is upward or downward.

```
import matplotlib.pyplot as plt
import random
prices = random.sample(range(100,150),10)
days = range(1,11)
plt.plot(days,prices, marker='o')
plt.title("Daily Stock Prices")
plt.xlabel("Days")
plt.ylabel("Stock Prices")
plt.show()
```



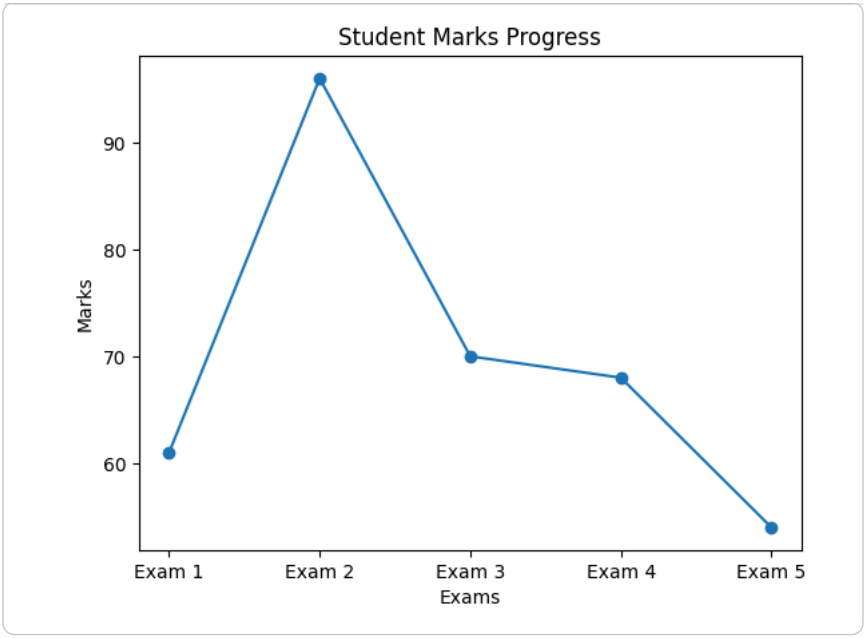
4. Plot a line chart to compare temperature changes of two cities over 7 days.

```
import matplotlib.pyplot as plt
import random
city1 = random.sample(range(30,40),7)
city2 = random.sample(range(30,40),7)
days = range(1,8)
plt.plot(days,city1, marker='o', label="City 1")
plt.plot(days,city2, marker='o', label="City 2")
plt.title("Temperature Changes")
plt.xlabel("Days")
plt.ylabel("Temperature")
plt.legend()
plt.show()
```



5. Create a line chart showing progress of a student's marks across 5 exams.

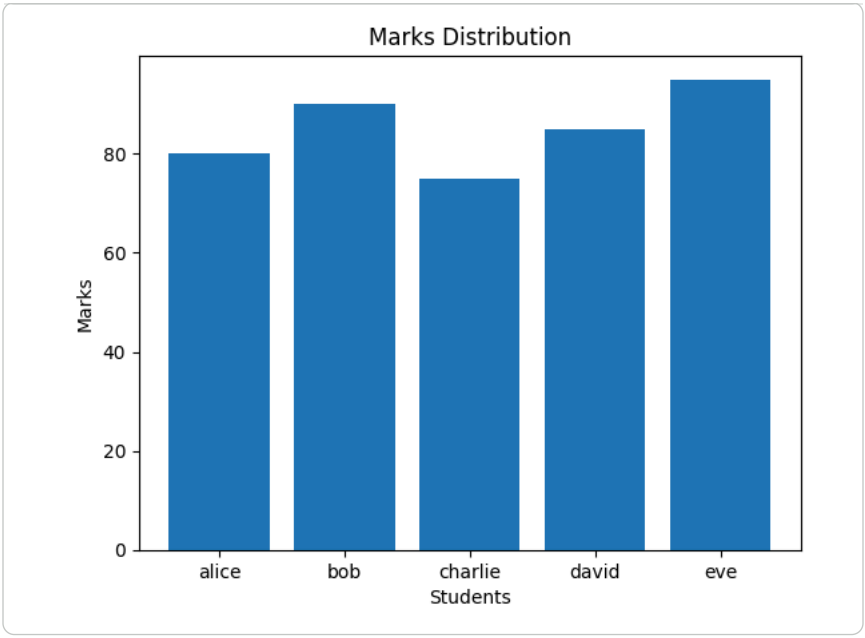
```
import matplotlib.pyplot as plt
import random
marks = random.sample(range(50,100),5)
exams = ["Exam 1", "Exam 2", "Exam 3", "Exam 4", "Exam 5"]
plt.plot(exams,marks, marker='o')
plt.title("Student Marks Progress")
plt.xlabel("Exams")
plt.ylabel("Marks")
plt.show()
```



C. Bar Chart Practice Questions

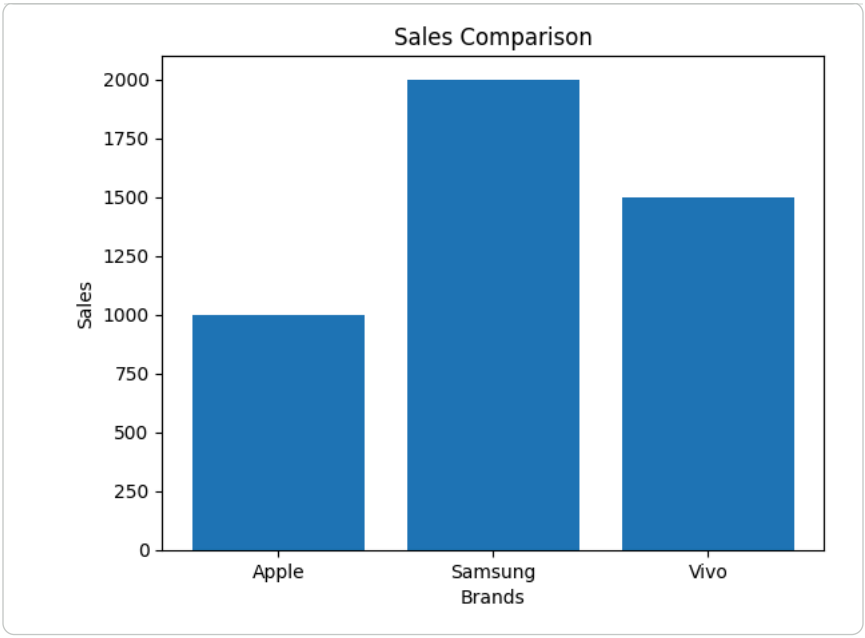
1. Create a bar chart showing marks scored by 5 students in an exam.

```
import matplotlib.pyplot as plt
marks = [80,90,75,85,95]
students = ["alice", "bob", "charlie", "david", "eve"]
plt.bar(students,marks)
plt.title("Marks Distribution")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```



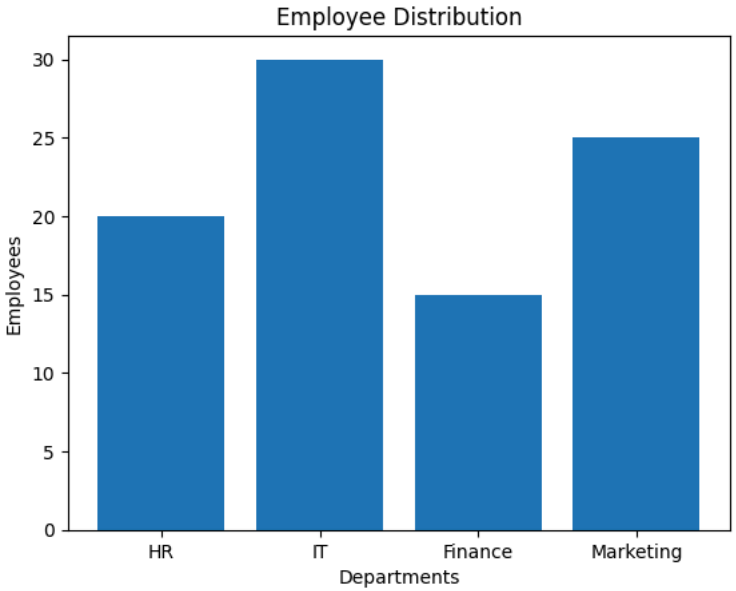
2. Plot a bar chart comparing sales of three different brands.

```
import matplotlib.pyplot as plt
brands = ["Apple", "Samsung", "Vivo"]
sales = [1000,2000,1500]
plt.bar(brands,sales)
plt.title("Sales Comparison")
plt.xlabel("Brands")
plt.ylabel("Sales")
plt.show()
```



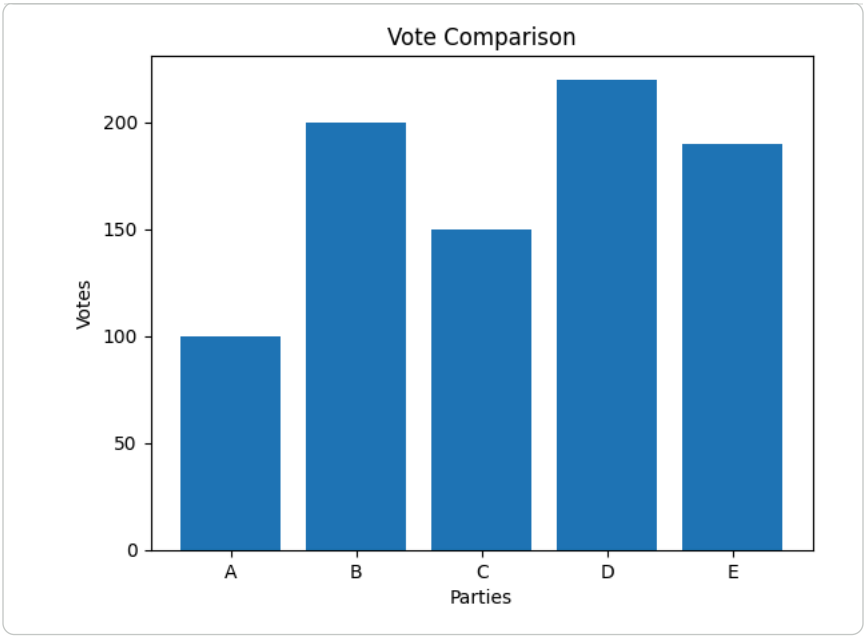
3. Draw a bar chart to represent number of employees in different departments.

```
import matplotlib.pyplot as plt
departments = ["HR", "IT", "Finance", "Marketing"]
employees = [20,30,15,25]
plt.bar(departments,employees)
plt.title("Employee Distribution")
plt.xlabel("Departments")
plt.ylabel("Employees")
plt.show()
```



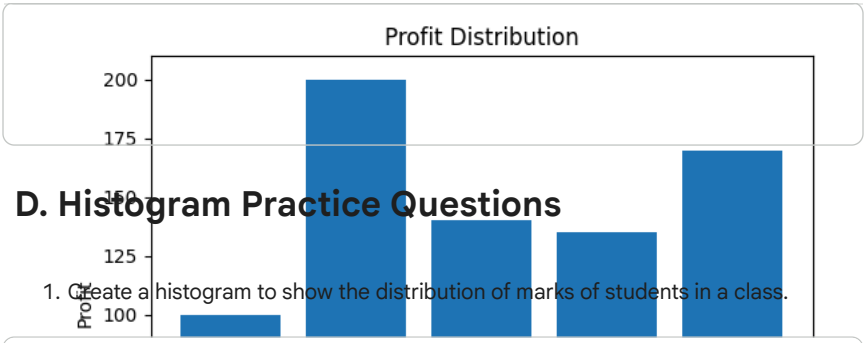
4. Create a bar chart to compare votes received by different political parties.

```
import matplotlib.pyplot as plt
parties = ["A", "B", "C", "D", "E"]
votes = [100,200,150,220,190]
plt.bar(parties,votes)
plt.title("Vote Comparison")
plt.xlabel("Parties")
plt.ylabel("Votes")
plt.show()
```



5. Plot a bar chart showing product-wise profit in a company.

```
import matplotlib.pyplot as plt
products = ["A", "B", "C", "D", "E"]
profit = [100,200,140,135,170]
plt.bar(products,profit)
plt.title("Profit Distribution")
plt.xlabel("Products")
plt.ylabel("Profit")
plt.show()
```



D. Histogram Practice Questions

1. Create a histogram to show the distribution of marks of students in a class.

```
import matplotlib.pyplot as plt
import random
marks = [random.randint(20, 100) for _ in range(100)]
plt.hist(marks, bins=10, edgecolor='black')
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.grid(axis='y', alpha=0.75)
plt.show()
```

